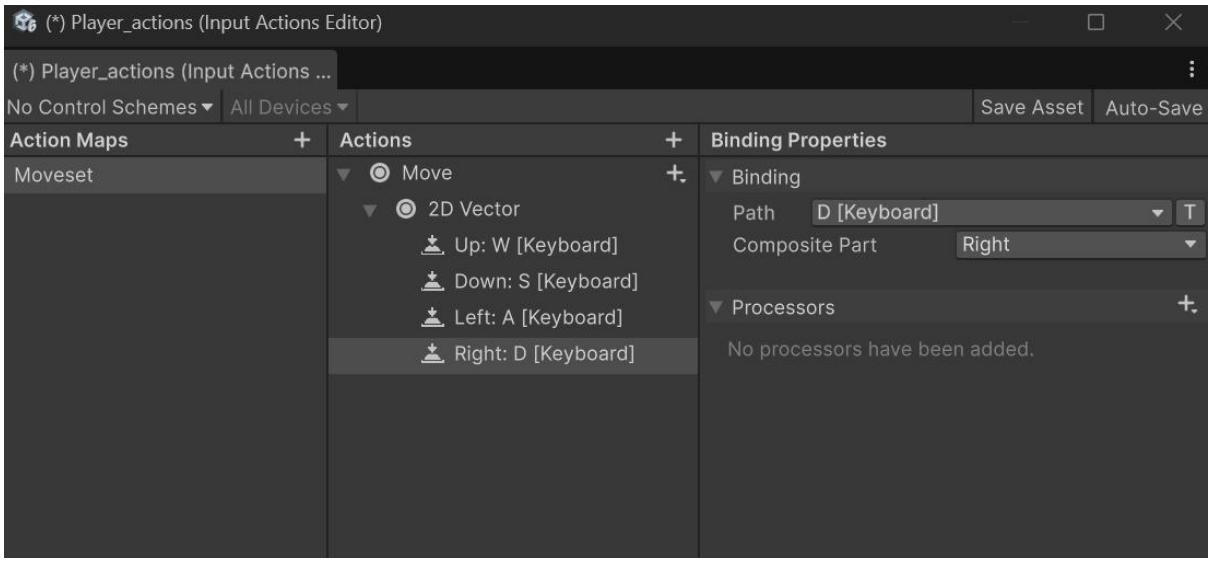
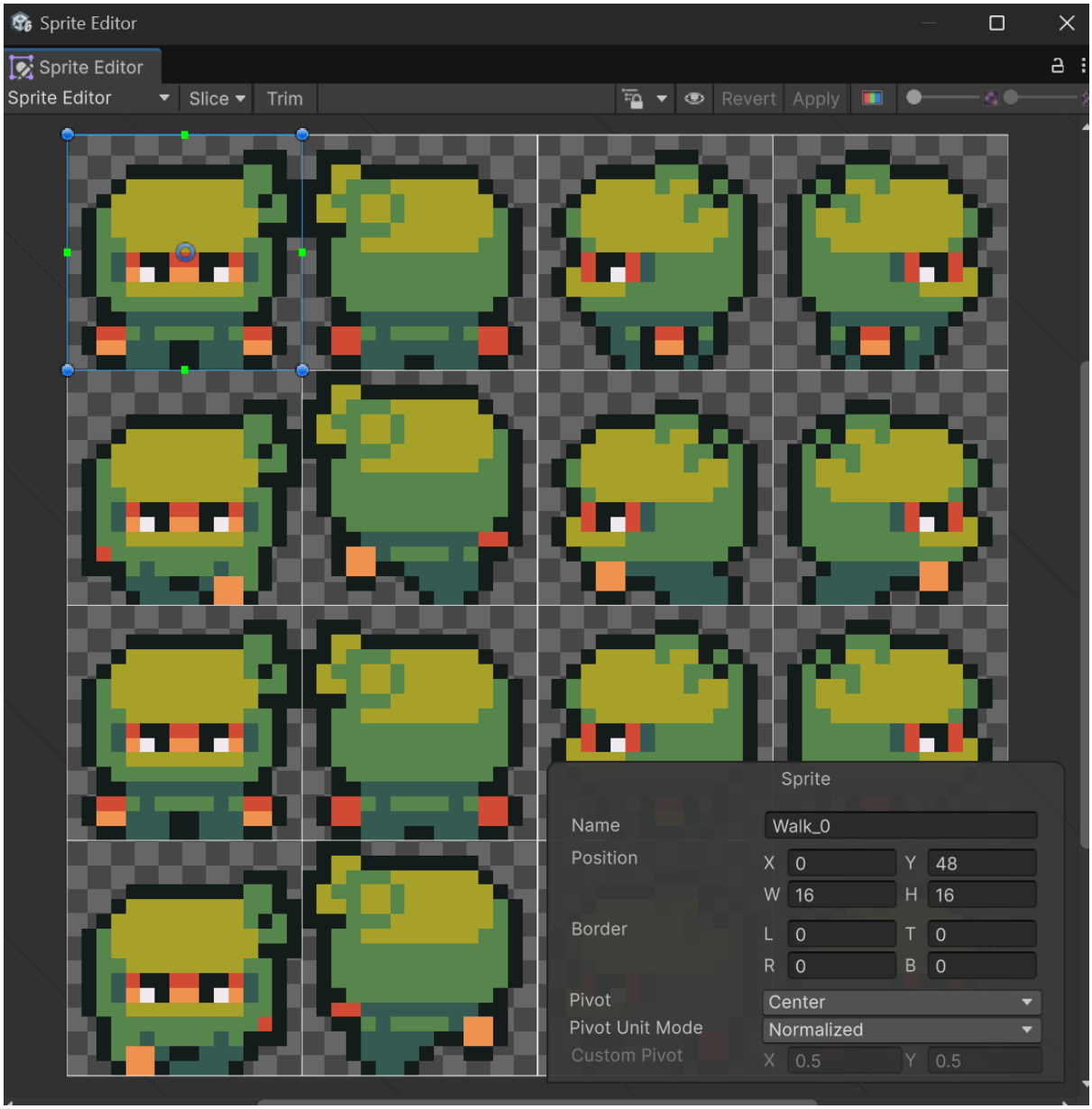
LAB 2

Level 1



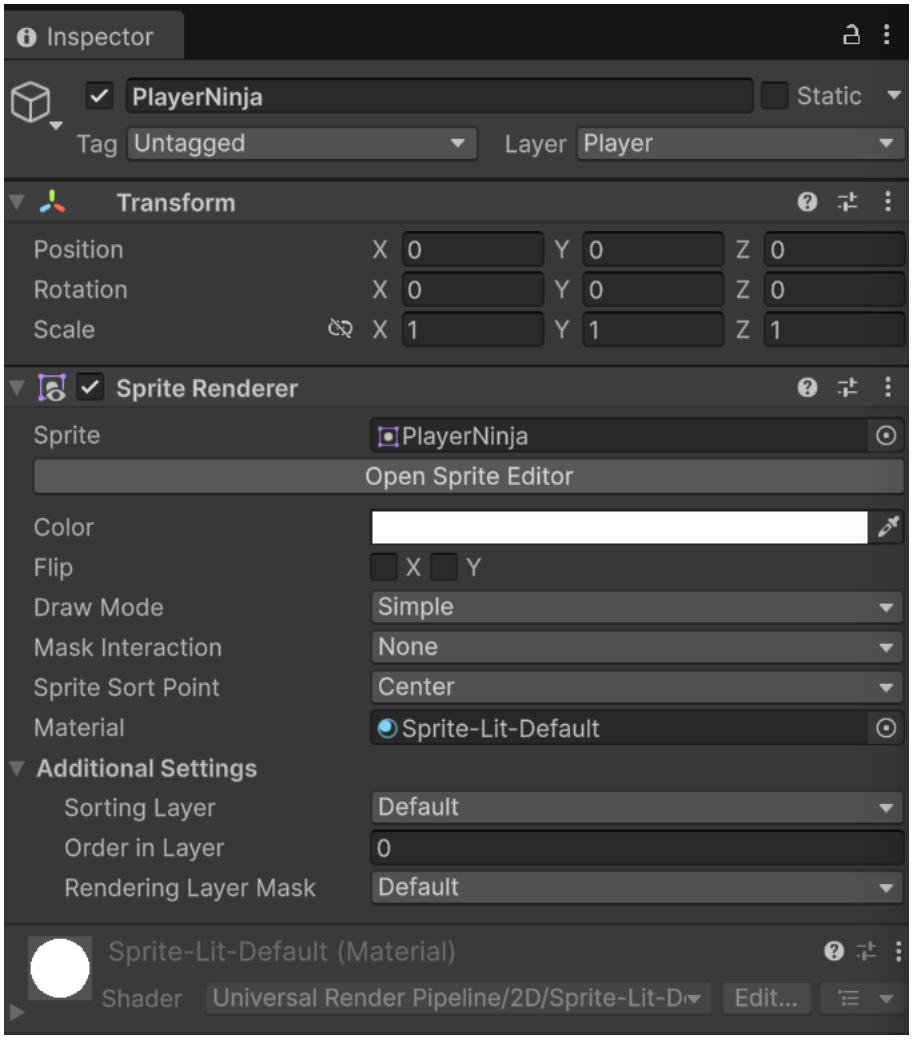
As it can be seen, the “Player_actions” allows use to use the action map to bind the directions of up down left and right, to the keys w s a d, which should in theory let the character move, but only after a few more levels of this report

Level 2

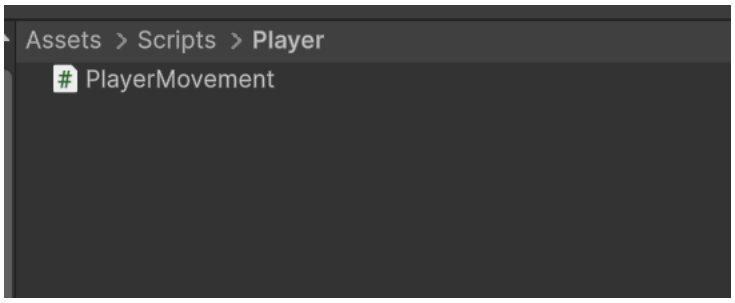


In this image you can see part of the “walk” sprite sheet, in which its possible to see that it’s made up of multiple 16 by 16 sprites, numbered Walk_0 through Walk_15, counted like if reading a book, start at the top left, go right, and when reaching the end you return to the left and go down a line

Level 3

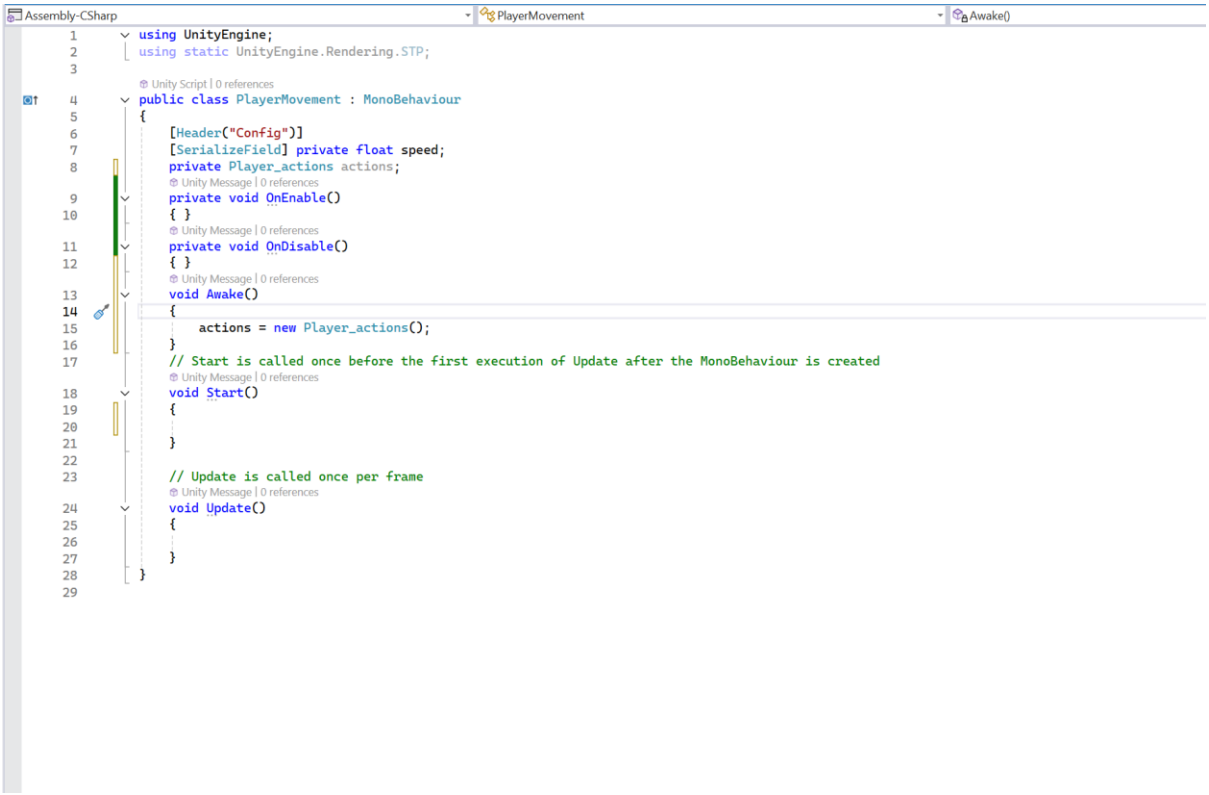


Now we have renamed our “Walk_0” to PlayerNinja, and added a new Layer called “Player” which we attributed to the Player Ninja



And we made a player movement script, which we added as a component to “PlayerNinja”

Level 4



In this code, we have a incomplete “translator” of the Input actions made before, and the player character, which should allow it to move, if the player object is able to

Level 5

```
using UnityEngine;
using static UnityEngine.Rendering.STP;

Unity Script | 0 references
public class PlayerMovement : MonoBehaviour
{
    [Header("Config")]
    [SerializeField] private float speed;
    private Player_actions actions;
    private Rigidbody2D rigid_var;
    private Vector2 moveDirection;
    Unity Message | 0 references
    private void OnEnable()
    { }
    Unity Message | 0 references
    private void OnDisable()
    { }
    Unity Message | 0 references
    void Awake()
    {
        GetComponent<Rigidbody2D>();
        actions = new Player_actions();
    }

    1 reference
    private void ReadMovement()
    {
        moveDirection = actions.Moveset.Move.ReadValue<Vector2>().normalized;
    }

    1 reference
    private void Move()
    {
rigid_var.MovePosition(rigid_var.position +
moveDirection * (speed * Time.fixedDeltaTime));
    }

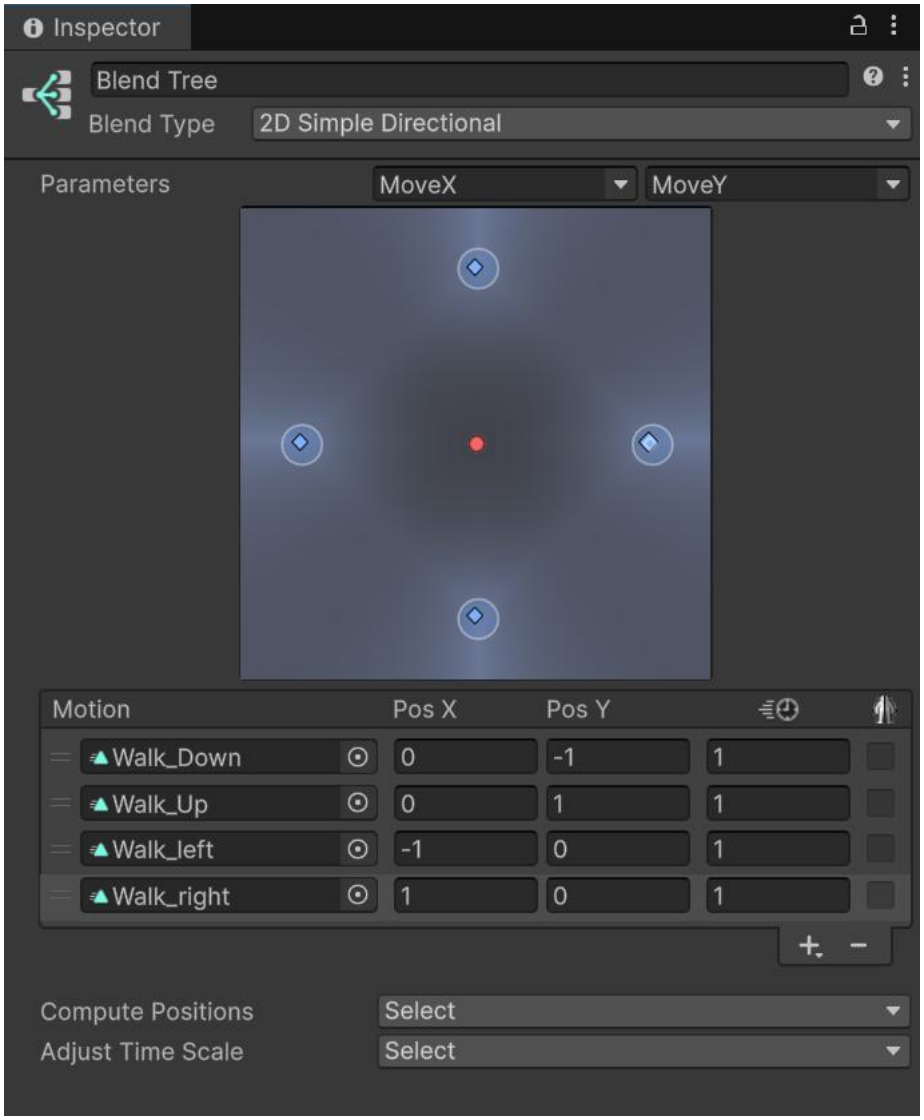
    Unity Message | 0 references
    private void FixedUpdate()
    {
        Move();
    }

    // Start is called once before the first execution of Update after the MonoBehaviour is created
    Unity Message | 0 references
    void Start()
    {

    }

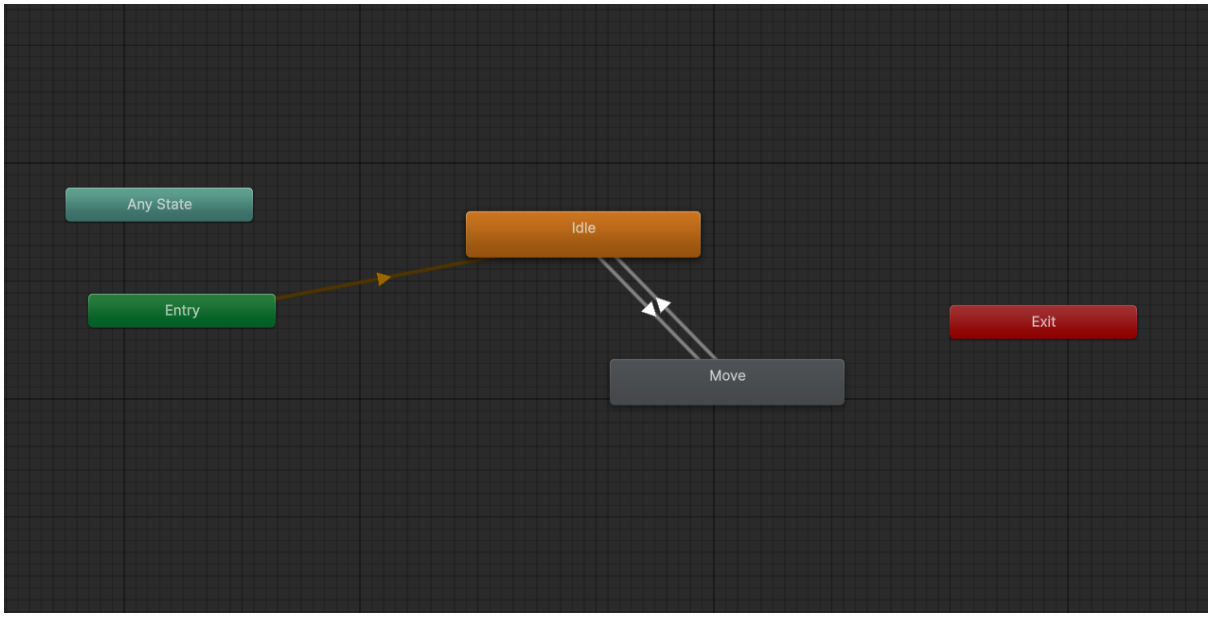
    // Update is called once per frame
    Unity Message | 0 references
    void Update()
    {
        ReadMovement();
    }
}
```

In level 5 we improved the work of level 4, by making it also consider velocity and the direction of the movement



Now we connect the animations to the movement itself, as that not only does the character move in a certain direction, it also looks like its moving

Bonus Level



And finally as part of the bonus level, we made an idle animation for the player character, along with making it switch between the idle and moving animations at will.

```
public class PlayerMovement : MonoBehaviour
{
    [Header("Config")]
    [SerializeField] private float speed;
    private Player_actions actions;
    private Rigidbody2D rigid_var;
    private Vector2 moveDirection;
    private Animator anim_ninja;

    Unity Message | 0 references
    void Awake()
    {
        rigid_var = GetComponent<Rigidbody2D>();
        actions = new Player_actions();
        anim_ninja = GetComponent<Animator>();
    }

    Unity Message | 0 references
    private void OnEnable()
    {
        actions.Enable();
    }
    Unity Message | 0 references
    private void OnDisable()
    {
        actions.Disable();
    }

    1 reference
    private void ReadMovement()
    {
        moveDirection = actions.Moveset.Move.ReadValue<Vector2>().normalized;
        if (moveDirection == Vector2.zero)
        {
            anim_ninja.SetBool("Moving", false);

            return;
        }
        anim_ninja.SetBool("Moving", true);
        anim_ninja.SetFloat("MoveX", moveDirection.x);
        anim_ninja.SetFloat("MoveY", moveDirection.y);
    }

    1 reference
    private void Move()
    {
    rigid_var.MovePosition(rigid_var.position + moveDirection * (speed * Time.fixedDeltaTime));
    }

    Unity Message | 0 references
    private void FixedUpdate()
    {
        Move();
    }

    // Start is called once before the first execution of Update after the MonoBehaviour is created
    Unity Message | 0 references
    void Start()
    {

    }

    // Update is called once per frame
    Unity Message | 0 references
    void Update()
    {
        ReadMovement();
    }
}
```

this comes along side some extra parts of the code, to change a Boolean variable between true or false, if the character is moving